



Counted Twice, Caught Once

What a human pill check still catches beyond the robot’s own fill log

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Background

Dispensing robots log their own weight-based fill accuracy for every prescription. Pharmacies also require a human pre-handover pill count on top. Little is known about how much of that second check’s value the first one hasn’t already banked.

Aims

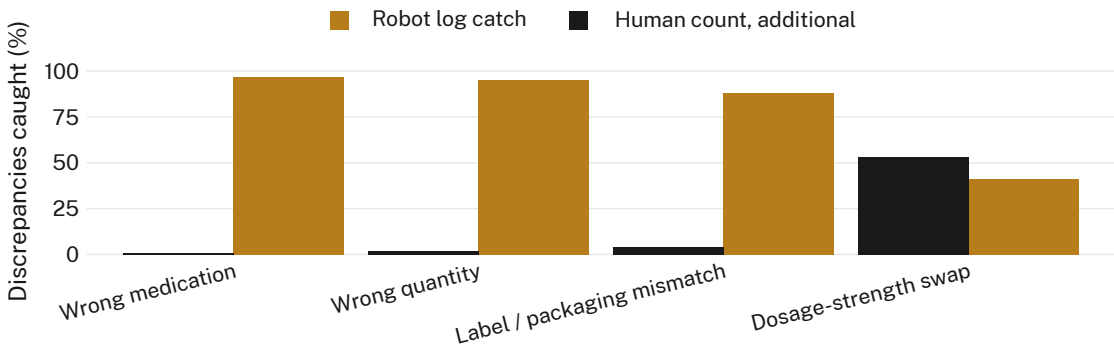
- Quantify how many additional discrepancies the human count catches beyond the robot’s own log
- Test whether independent catches concentrate in a specific error type
- Compare that concentration against a chain policy applying the same check at every handover regardless

Methods

- 42 pharmacy sites · merged robot-log and human-count records · 10-month window (Apr 2024–Jan 2025)
- 2,918 flagged discrepancies classified by error type; catch attributed to whichever check logged it first
- Site staffing and count protocol held constant across the window

Results

The human count independently catches an additional 6.1% of flagged discrepancies overall; 71% of those additional catches sit in one category the robot’s own log records as a correct fill.



Independent human-count catches by error type, 42 pharmacy sites, April 2024–January 2025 (n = 2,918 flagged discrepancies). Dosage-strength swaps account for the large majority of catches the robot’s own log would otherwise have missed.

Discussion & conclusions

A blanket verification layer applied uniformly returns almost all of its marginal value from one narrow, identifiable failure class; elsewhere it mostly reconfirms what the log already knows. Next: a count directed only at dosage-strength-ambiguous fills, rather than every handover.

References

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De-identified fill logs only; no prescriber or patient identifiers retained.

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“The log says correct. The count disagrees — in one place only.”

Independent catches concentrate in a single dosage class across 42 pharmacy sites.